Outsourcing Critical Thinking?
AI and the Production of Learning in Higher Education*

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Abstract

We examine how generative AI affects the time and cognitive effort students allocate to learning activities—inputs to human capital formation. Using a novel survey of college students, we measure time and effort across academic tasks with and without AI assistance, as well as how students reallocate time freed by AI. Students adopt AI strategically, concentrating use on time-intensive, independent tasks rather than on interactive tasks involving instructors or peers. AI reduces both time spent and cognitive effort, and these reductions are positively correlated—suggesting that time savings come bundled with reduced cognitive engagement. Students reallocate nearly all time saved by AI to leisure, sleep, and social activities rather than to additional academic work. Taken together, these patterns suggest that AI may reduce human capital accumulation by lowering both the time and effort students devote to learning.

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1 Introduction

College students now have access to technology that can write their essays, solve their problem sets, and provide personalized tutoring at instant speed and near-zero marginal cost. The rapid adoption of generative AI tools like ChatGPT has sparked intense debate about their impact on higher education. Many observers worry that students may delegate critical thinking to these systems rather than developing skills themselves, with consequent reductions in learning and human capital formation (e.g., Berg, 2025; Chen, 2025; Clune, 2025; O'Rourke, 2025; Weatherby, 2025).

Understanding how AI affects learning requires measuring its impact on the inputs to the learning production process—specifically, students' allocation of time and cognitive effort. If students save time by using AI, whether this improves or harms learning depends on whether the time savings come from activities that are conducive to learning, and whether freed time is reallocated to more or less productive uses. Similarly, if AI substitutes for cognitive effort in tasks where struggle itself builds skills, its widespread use could reduce human capital accumulation even as it raises short-run academic performance.

To study these dynamics systematically, we adapt the task-based framework (Acemoglu and Autor, 2011; Autor and Dorn, 2013). The canonical approach decomposes occupations into bundles of tasks and studies how technology affects employment by changing which tasks workers perform. We apply this framework to education by decomposing learning into specific activities—attending lectures, solving problem sets, writing essays—and analyzing how students allocate time and cognitive effort across them. But education differs from standard production settings in a crucial way: the outcome is inseparable from the process. In manufacturing, it does not matter whether robots or humans assembled a car, only that the car functions properly. In learning, the cognitive effort exerted while struggling through a problem set generates understanding that cannot be replicated by simply receiving the answer. This inseparability creates a fundamental tension when AI automates academic work: it may eliminate the very mechanism through which human capital accumulates.

The model delivers two main insights. First, whether AI increases or decreases learning within a given activity depends on the balance of three forces: the *time effect* captures whether AI compresses or expands the time students spend on the activity; the *effort effect* captures whether AI substitutes for cognitive effort, reducing engagement per unit of time; and the *productivity effect* captures whether AI makes each unit of effort more productive through feedback or personalized instruction. Second, the impact of AI on overall learning

depends on a previously neglected factor: how students reallocate time freed by AI across activities. Under optimal time allocation, aggregate effects depend only on productivity-weighted gains across activities—a Hulten-style envelope theorem. But students may not allocate time in way that maximizes learning. Suppose a student saves three hours by using ChatGPT to write an essay. Whether this increases or decreases aggregate learning depends on where those three hours go—to problem sets that build foundational skills, in which case learning may rise, or to video games, in which case learning may fall. This reallocation margin is absent from experimental studies, yet it may dominate the direct effects of AI on any single activity.

Shedding light on these empirical relations requires data on how students allocate time and effort across specific learning activities—data that existing surveys do not provide. We address this gap by conducting a novel survey that measures how college students allocate time and effort across academic tasks, and how this allocation has changed with the adoption of generative AI. Our approach builds on the education production function framework used to study the relationship between student time use and academic performance (Dolton et al., 2003; Stinebrickner and Stinebrickner, 2004). We extend this framework to account for the possibility that AI changes both the productivity of time spent on each task and students' decisions about which tasks to prioritize.

Our survey reveals three main findings. First, AI adoption varies dramatically across academic tasks. Students adopt AI primarily for independent work—readings, problem sets, writing, and exam preparation—rather than for tasks involving direct interaction with instructors or peers, such as office hours or campus resources. Adoption is higher for tasks that students rate as important for grades and for tasks that consume more time, suggesting that students deploy AI strategically on consequential, time-intensive tasks rather than indiscriminately.

Second, AI reduces time spent on nearly all academic tasks. Students report spending roughly 16 percent less time on tasks when using AI. Time savings are largest for tasks on which students already spend the most time, and vary substantially across students: STEM majors save nearly twice as much time as non-STEM majors, and students with above-median SAT scores save more than double the time of those below the median. Students reallocate nearly all time saved by AI to non-academic activities. Among the 83 percent of AI users who report that AI saves them time—roughly five hours per week on average—leisure absorbs the largest share (30 percent), followed by sleep (20 percent) and social activities (17 percent). Only 7 percent of freed time flows to other academic work.

Third, AI reduces cognitive effort across nearly all academic tasks. Students report exerting roughly 11 percent less cognitive effort with AI than they would without it. Unlike time savings, effort reductions are remarkably uniform across student characteristics. The pattern of effort reduction varies across types of thinking. Using six cognitive categories based on Bloom's taxonomy, we find that AI reduces effort most consistently for *understanding* and *synthesizing*—summarizing, explaining, and integrating information—with 37 percent of task observations showing reduced effort. Effects are more modest at both extremes of the taxonomy: 29 percent report reduced effort for *remembering* facts, while *evaluating* arguments shows the smallest share reporting reduced effort (24 percent) and the largest share reporting *increased* effort (38 percent).

Importantly, time savings and effort reductions are positively correlated—AI saves time precisely by reducing the cognitive work required to complete a task. This suggests that the benefits of AI (time savings) come bundled with costs (reduced cognitive engagement). Whether this trade-off harms learning depends on whether freed time is reinvested productively; our reallocation results suggest it is not. Moreover, the benefits are immediate while the costs—in the form of reduced learning—are delayed, which may lead students to overweight present gains. Consistent with this, 31 percent of students report using AI more than they believe they should, a pattern that points to a self-control problem.

Our paper contributes to a nascent literature on how AI affects human capital accumulation. Existing work follows two broad approaches. One strand uses randomized experiments that assign access to AI tools and measure subsequent performance on specific tasks (e.g., Lehmann et al., 2024; Bastani et al., 2025; Lira et al., 2025; Kreijkes et al., 2026). These studies offer strong internal validity but limited external validity: controlled settings may differ substantially from how students naturally interact with AI, and outcomes measured are specific to experimental tasks. A second strand provides descriptive evidence on AI adoption and usage patterns among students (e.g., Hirabayashi et al., 2024; Stöhr et al., 2024; Ravšelj et al., 2025). This work documents that students use AI for activities that may complement cognitive effort (such as explaining concepts) and others that may substitute for it (such as generating essay text). But without measuring how AI affects the time and effort students allocate to these activities, descriptive work alone cannot determine whether observed usage patterns increase or decrease learning.

Our paper bridges these approaches by combining naturalistic measurement of how students use AI across academic activities with a theoretical framework that connects these inputs to learning outcomes. The task-based model provides structure for interpreting

survey responses in terms of their implications for human capital accumulation, while the survey design captures AI use as it occurs in students' actual academic lives.

We also contribute to the literature on technology and learning. A substantial body of work documents the importance of student time allocation for academic outcomes (e.g., Dolton et al., 2003; Stinebrickner and Stinebrickner, 2004; Babcock and Marks, 2011), but relatively little is known about how productivity-enhancing technologies alter the way students allocate time and effort across learning activities. We contribute by providing the first comprehensive evidence on how generative AI reshapes students' time and effort allocation, and by developing a framework that connects these changes to aggregate learning outcomes.

The remainder of the paper proceeds as follows. Section 2 develops a task-based model of learning production. Section 3 describes the survey design. Section 4 presents the data and empirical approach. Section 5 presents the main findings. Section 6 concludes.

2 A Task-Based Model of Learning Production

2.1 Baseline Model Without AI

A student allocates time across N learning activities indexed by $i \in \{1, \ldots, N\}$ and a composite "other" activity (or outside option) indexed by $i = 0$. Learning activities include attending lectures, solving problem sets, reading textbooks, participating in discussions, visiting office hours, and writing essays. The other activity encompasses non-academic pursuits including leisure, sleep, exercise, paid work, extracurricular activities, and socializing.

Each learning activity $i \in \{1, \ldots, N\}$ is characterized by two choice variables and one parameter: time allocated to activity i, denoted $t_i \geq 0$ and measured in hours; cognitive effort intensity, denoted $m_i \in [0, 1]$, which measures effort per clock hour; and the productivity of effort, denoted $\phi_i > 0$, which determines learning output per unit of effort-hour.[1]

Time allocated to the other activity is denoted $t_0 \geq 0$. This activity does not require cognitive effort in the same sense as learning activities, so we set $m_0 = 0$ by convention. However, it may affect learning indirectly through rest, stress reduction, physical health, or skill development in non-academic domains.

[1] Activities with low ϕ_i may include reading a textbook that is poorly matched to the student's preparation level—either too advanced to be comprehensible or too elementary to be informative—attending disorganized lectures, or completing tasks with high administrative overhead relative to intellectual content, such as formatting problem sets or navigating course management software.

Learning produced from activity $i \in \{1, \ldots, N\}$ is given by a Cobb-Douglas production function:

$$y_i = \phi_i \, t_i^{\alpha_i} \, m_i^{\beta_i}, \qquad \alpha_i, \beta_i > 0. \tag{1}$$

The parameters α_i and β_i capture the curvature of the learning production function in time and intensity, respectively. When $\alpha_i < 1$, there are decreasing returns to time in activity i (e.g., the tenth hour of reading may produce less learning than the first hour). When $\beta_i < 1$, there are decreasing returns to cognitive intensity: doubling mental effort does not double learning.

The Cobb-Douglas specification in equation (1) implies that time and cognitive effort are complements in producing learning. Spending time on an activity without mental engagement ($m_i \approx 0$) produces little learning, just as high mental engagement yields limited benefits without sufficient time to apply it.

The other activity affects learning indirectly. Let $g(t_0)$ capture the effect of time in the other activity on learning. For tractability, we assume $g(t_0) = \theta t_0^{\delta}$ with $\delta \in (0, 1)$ and $\theta > 0$, capturing decreasing returns to time in the other activity. The parameter θ determines the importance of non-academic time for learning, with higher θ indicating that other activities are more complementary to learning.

Total learning is an aggregator function over activity-specific learning, scaled by the effect of non-academic time:

$$L = g(t_0) \cdot f(y_1, \ldots, y_N). \tag{2}$$

For tractability, we assume f is a CES aggregator:

$$f(y_1, \ldots, y_N) = \left(\sum_{i=1}^{N} \omega_i y_i^{\rho} \right)^{1/\rho}, \qquad \rho \leq 1, \tag{3}$$

where $\omega_i > 0$ captures the importance of activity i for aggregate learning—reflecting, for example, how central the activity is to the curriculum or how foundational the skills it develops are for subsequent learning—and $\sigma \equiv 1/(1-\rho)$ is the elasticity of substitution between learning activities. The elasticity of substitution plays a central role in determining AI's aggregate effects. When $\sigma < 1$, learning activities are strong complements, and AI-induced declines in one activity are difficult to compensate for by increasing another. When

$\sigma > 1$, students can substitute across activities more readily.

2.2 Introducing AI

AI enters as an activity-specific input. Let $k_i \geq 0$ denote the intensity of AI use in learning activity i. For example, k_i might represent whether a student has access to an AI tutor like ChatGPT or Claude while working on problem sets, or whether they use AI to summarize readings. We assume AI does not directly affect the other activity, so $k_0 = 0$.

We model AI as affecting all three components of learning production. Specifically:

$$t_i(k_i) = t_i^0 \cdot \tau_i(k_i), \qquad m_i(k_i) = m_i^0 \cdot \mu_i(k_i), \qquad \phi_i(k_i) = \phi_i^0 \cdot \pi_i(k_i), \tag{4}$$

where t_i^0, m_i^0, ϕ_i^0 denote baseline values without AI, and $\tau_i(\cdot), \mu_i(\cdot), \pi_i(\cdot)$ are shifter functions with $\tau_i(0) = \mu_i(0) = \pi_i(0) = 1$.

The shifter functions capture three distinct mechanisms. The function $\tau_i(k_i)$ represents the time effect, with $\tau_i < 1$ indicating that AI reduces the time needed for activity i and $\tau_i > 1$ indicating that AI increases time spent. The function $\mu_i(k_i)$ captures intensity substitution: $\mu_i < 1$ indicates that AI reduces cognitive effort per hour, whereas $\mu_i > 1$ indicates that AI increases intensity. The function $\pi_i(k_i)$ captures effectiveness: $\pi_i > 1$ indicates that AI makes effort more productive, whereas $\pi_i < 1$ indicates that AI reduces effectiveness.

Learning from activity i under AI is:

$$\begin{aligned} y_i(k_i) &= \phi_i(k_i)\, t_i(k_i)^{\alpha_i}\, m_i(k_i)^{\beta_i} \\ &= \phi_i^0 \pi_i(k_i) \left[t_i^0 \tau_i(k_i)\right]^{\alpha_i} \left[m_i^0 \mu_i(k_i)\right]^{\beta_i}. \end{aligned} \tag{5}$$

Taking logs and differentiating with respect to k_i yields a decomposition of how AI affects learning in activity i:

$$\frac{d \ln y_i}{dk_i} = \underbrace{\frac{d \ln \pi_i}{dk_i}}_{\text{effectiveness}} + \alpha_i \underbrace{\frac{d \ln \tau_i}{dk_i}}_{\text{time compression}} + \beta_i \underbrace{\frac{d \ln \mu_i}{dk_i}}_{\text{mental effort}}. \tag{6}$$

This decomposition shows that AI's effect on learning in activity i operates through three channels. The effectiveness effect captures whether AI makes each unit of effort more productive. The time compression effect captures whether students spend more or

less time on the activity. The effort effect captures whether students exert more or less cognitive effort per unit of time.

2.3 Reallocation and Aggregate Effects

The activity-level decomposition in equation (6) considers each activity in isolation, without accounting for how time freed from one activity may be reallocated to others. When AI compresses time in some learning activities ($\tau_i < 1$), this relaxes the time constraint and allows reallocation. Crucially, freed time can be reallocated either to other learning activities or to the outside option. The reallocation pattern plays a critical role in determining whether AI increases or decreases aggregate learning.

To formalize the aggregate effects, define the learning-share weight for activity i as:

$$s_i \equiv \frac{\partial \ln f}{\partial \ln y_i} = \frac{\omega_i y_i^\rho}{\sum_{j=1}^N \omega_j y_j^\rho}, \qquad \sum_{i=1}^N s_i = 1. \tag{7}$$

These weights capture each activity's contribution to aggregate learning, encoding both the importance parameters ω_i and the current composition of learning through y_i. Activities that produce more learning receive higher weight.

Since $L = g(t_0) \cdot f(y_1, \ldots, y_N)$, a first-order change in aggregate learning satisfies:

$$d \ln L = \delta \, d \ln t_0 + \sum_{i=1}^N s_i \, d \ln y_i. \tag{8}$$

Aggregate learning changes as a weighted average of activity-level learning changes, plus the effect of the outside option. This is the direct analog of Hulten's theorem in production theory.

Substituting the activity-level decomposition from equation (6) into equation (8) yields the full expression for how AI in activity i affects aggregate learning:

$$\frac{\partial \ln L}{\partial k_i} = \underbrace{s_i \left(\frac{d \ln \pi_i}{dk_i} + \alpha_i \frac{d \ln \tau_i}{dk_i} + \beta_i \frac{d \ln \mu_i}{dk_i} \right)}_{\text{direct effect on activity } i} + \underbrace{\delta \frac{\partial \ln t_0}{\partial k_i}}_{\text{outside option}} + \underbrace{\sum_{j \neq i} s_j \left(\alpha_j \frac{\partial \ln t_j}{\partial k_i} + \beta_j \frac{\partial \ln m_j}{\partial k_i} \right)}_{\text{reallocation spillovers}}. \tag{9}$$

The first term captures the direct effect of AI on activity i, weighted by its learning share. The second term captures how AI-induced time savings in activity i affect time allocated

to the outside option. The third term captures reallocation spillovers: how time and effort in other learning activities $j \neq i$ respond to AI use in activity i.

Under optimal time allocation, a version of Hulten's theorem applies. At an interior optimum, students equate the marginal return to time across all activities, and the envelope theorem implies that reallocation effects vanish to first order. Formally, if students solve $\max L$ subject to a time constraint, the first-order conditions ensure that $\partial \ln t_0 / \partial k_i$ and $\partial \ln t_j / \partial k_i$ adjust to maintain optimality, and these adjustments have no first-order effect on L. The aggregate effect of AI then simplifies to:

$$\frac{\partial \ln L}{\partial k_i} \approx s_i \left(\frac{d \ln \pi_i}{dk_i} + \alpha_i \frac{d \ln \tau_i}{dk_i} + \beta_i \frac{d \ln \mu_i}{dk_i} \right). \tag{10}$$

In this benchmark, aggregate learning responds as the share-weighted sum of activity-level effects, and the reallocation terms in equation (9) do not appear.

However, this logic breaks down when students do not allocate time optimally. Suppose AI compresses time in a subset of learning activities $\mathcal{A}_{\text{AI}} \subset \{1, \ldots, N\}$, where $\tau_i < 1$ for all $i \in \mathcal{A}_{\text{AI}}$. The time saved is $\Delta T = \sum_{i \in \mathcal{A}_{\text{AI}}} t_i^0 (1 - \tau_i)$. We parameterize the reallocation decision by letting $\eta \in [0, 1]$ denote the share of freed time reallocated to learning activities, so $(1 - \eta)$ is reallocated to the outside option. The parameter η is a reduced-form summary of student behavior rather than a quantity derived from optimization. If $\eta < 1$, some freed time flows to the outside option even when the marginal return to other learning activities is higher, and the reallocation terms in equation (9) no longer vanish.

Importantly, reallocation to the outside option is not necessarily harmful for learning. The outside option includes rest, and when students are sleep-deprived or overstressed, increasing t_0 may raise the productivity of remaining study hours through $g(t_0)$. The net effect depends on whether the marginal return to rest exceeds the marginal return to additional study time—a comparison that the Hulten benchmark assumes students make optimally, but that behavioral frictions may distort.

The model therefore implies that AI's aggregate effect on learning depends on the balance of four forces: productivity gains, time compression, and effort substitution within each activity (the direct effects in equation 10), plus reallocation across activities (the additional terms in equation 9). Measuring AI's aggregate effect requires estimating all four components—data that existing surveys do not provide. In the next section, we describe a survey designed to measure these quantities among college students.

3 Survey Design

The model implies that measuring AI's effect on learning requires data on three quantities for each learning activity: time allocation (t_i and τ_i), cognitive effort intensity (m_i and μ_i), and productivity (ϕ_i and π_i). Measuring aggregate effects additionally requires data on how students reallocate time freed by AI across activities (η). We designed a survey to collect these measures from college students.

3.1 Survey Structure

The survey consists of three main components: demographic questions, questions about academic tasks, and questions about general AI usage patterns.

Participants begin by providing demographic information including age, gender, race and ethnicity, international student status, high school type (public or private), and first-generation college student status. We collect academic background measures including standardized test scores (SAT and ACT, where applicable), current GPA, institution name, year in program, and major or primary field of study. Participants also report the total hours they typically spend on academic activities in a given week. We also measure self-reported intrinsic motivation to learn and time preferences (Falk et al., 2023) to examine if these characteristics moderate AI's effects on learning.

3.2 Task-Level Measurement

The core of the survey measures time allocation and cognitive effort across academic tasks. We developed a list of 13 common academic activities spanning content consumption (attending lectures, completing readings, watching recorded lectures), active learning (note-taking, exam preparation, problem-solving), knowledge production (writing papers, revising written work evaluating sources), and collaboration and support-seeking (group projects, office hours, campus resources, hands-on activities). Appendix Table A1 describes each activity in more detail. Each participant responds to questions about three randomly selected tasks from this list. For each selected activity, we collect measures corresponding to the model's key parameters:

Task characteristics. Participants rate each task's importance on a scale from 0 (not at all important) to 10 (extremely important) across four dimensions: contribution to learning, contribution to course grades, relevance to skills needed after college, and personal enjoyment or engagement. These measures help characterize the perceived importance of

the activity for aggregate learning (ω_i in equation 3). Participants also report typical weekly time spent on the task (t_i). We measure cognitive effort using a scale from 0 ("I do this task on autopilot without really thinking") to 10 ("I concentrate intensely and think carefully the whole time"), corresponding to m_i in the model. The instructions emphasize that this scale captures intensity of mental effort rather than time spent.

AI use. Participants report how frequently they use generative AI tools (such as ChatGPT, Claude, or Gemini) for each task using a five-point scale: always or almost always, often (more than half the time), sometimes (less than half the time), rarely, or never. This measures AI intensity k_i. Participants who report using AI at least rarely describe in their own words how they typically use AI for the task, including what they are trying to accomplish and how they interact with the tool. Participants who never use AI for the task explain their reasons. These open-ended responses provide qualitative evidence on the mechanisms through which AI affects learning inputs.

Counterfactual time and effort. To measure how AI affects time allocation (τ_i) and cognitive effort (μ_i), AI users estimate both the time required and the mental effort exerted to complete each task with and without AI assistance. The survey displays participants' reported time and effort levels with AI, then asks them to estimate how much time they would spend and how much mental effort they would exert if they completed the task entirely without AI tools. We use these measure to construct estimates of τ_i and μ_i for each activity.

Cognitive effort by type. Before measuring how AI affects different types of cognitive effort, we introduce participants to six categories based on Bloom's taxonomy: remembering, which involves recalling facts, terms, or information; understanding, which involves summarizing, explaining, or describing in one's own words; problem solving, which involves using knowledge to solve problems or complete tasks; analyzing, which involves breaking down information, examining relationships, and identifying patterns; synthesizing, which involves combining ideas from multiple sources to form new insights; and evaluating, which involves judging quality, validity, or effectiveness.

Participants who use AI for a given task then rate whether AI requires more or less mental effort compared to completing the activity without AI, separately for each cognitive category. The response scale includes five options ranging from much less mental effort to much more mental effort (plus a "not applicable" option). This disaggregated measurement allows us to examine whether AI substitutes for lower-order cognitive activities (remembering, understanding) while preserving or enhancing higher-order activities

(analyzing, synthesizing, evaluating).

Perceived optimal AI use. Finally, for each task, AI users indicate whether they feel they currently use AI the right amount, or whether they would prefer to use it more or less. This measure captures potential gaps between actual and desired AI use that may reflect self-control problems.

3.3 Time Reallocation

The survey next measures time reallocation patterns at the aggregate level. Participants who use AI for academic work first indicate whether AI saves them time overall, adds time overall, or makes no difference to their total academic work time. This categorical measure captures net time effects across all activities, not just the three randomly selected tasks.

Participants who report that AI saves them time estimate the weekly time savings and then allocate this freed time across nine categories using percentage sliders that must sum to 100 percent: paid work, extracurricular activities, social activities, entertainment and leisure, personal care, sleeping, commuting and transportation, other academic activities, and other. The share allocated to other academic activities provides a direct measure of η. Participants who report that AI adds time estimate the additional hours and indicate where this time comes from using the same category structure, allowing us to examine which activities students sacrifice when AI increases their academic workload.

3.4 General AI Usage

The survey concludes with questions about general AI usage patterns that extend beyond the three randomly selected tasks. Participants report how often they use generative AI during the academic semester on a five-point scale ranging from never to very frequently (daily or almost daily). Those who use AI report which specific tools they use regularly (including free and paid versions of ChatGPT, Claude, Gemini, Grok, and NotebookLM) and their monthly spending on AI subscriptions.

We measure perceived effects of AI on learning, grades, skill development, and course enjoyment using a five-point scale ranging from "significantly improves" to "significantly reduces." The survey also elicits beliefs about what percentage of college students use AI for schoolwork, for leisure, and for any purpose. Finally, an open-ended question asks participants to describe how AI has affected their critical thinking skills and time allocation.

4 Data: Novel Student Survey

4.1 Data Collection and Sample

We collected data on college students' time allocation, cognitive effort, and AI usage through an original survey administered via Prolific—an online platform widely used in social science research for recruiting survey participants (Palan and Schitter, 2018). We restricted eligibility to individuals currently enrolled as full-time undergraduates in the United States. Unlike previous surveys of student AI use (e.g., Lee et al., 2025), we did not condition participation on prior AI experience, which would introduce selection bias into our estimates of AI's effects on time and effort allocation. The survey took approximately 16 minutes to complete. The full survey instrument is available in Supplementary Materials.

Throughout the analysis, we report results at two levels of aggregation. In the student-level analyses, each observation represents one individual. In the task-level analyses, each observation represents one student-task combination. We pool across all task observations, clustering standard errors at the student level to account for the fact that each participant responds to questions about three randomly selected tasks.

4.2 Descriptive Statistics

Table 1 presents student-level summary statistics. We recruited 297 undergraduates, of whom 257 (87 percent) reported using generative AI for academic purposes during the current term. The average respondent is 21 years old, has a GPA of 3.47, and reports spending 26 hours per week on academics. The sample is roughly balanced by gender and spans a range of fields, with 39 percent in STEM, 32 percent in social sciences, and 11 percent in humanities. The majority attended public high schools (89 percent), and 35 percent are first-generation college students. Among AI users, 53 percent use AI frequently or very frequently during the term, and 83 percent report that AI saves them time on academic work.

Table 2 presents task-level summary statistics. There is substantial variation in time allocation across tasks (see also Appendix Figure A1). Students spend an average of 11 hours per week attending lectures but only 1 hour visiting office hours, with intermediate values for activities like problem sets (8 hours) and writing (5 hours). There is also significant variation in perceived importance (see also Appendix Figure A2). Students rate

lectures, problem sets, and exam preparation as most important for learning and grades, while activities like campus resources and group projects rank lower. Notably, the importance rankings differ across dimensions: problem sets rank high for learning and grades but low for enjoyment, whereas labs rank high on all four dimensions.

5 AI and the Production of Knowledge

5.1 AI Adoption Across Academic Tasks

We begin by documenting which academic tasks students use AI for and how adoption varies with task characteristics. Figure 1 reports the share of AI users who use AI for each task, broken down by intensity: rarely (one to two times out of ten), sometimes (three to five times), often (six to eight times), or always (nine to ten times). Figure 2 examines whether AI adoption is related to task characteristics. Each panel plots the share of students using AI against a different measure of how students perceive the task.

AI adoption varies dramatically across tasks. Figure 1 shows that over three-quarters of students use AI for readings (78 percent), problem sets (77 percent), writing (77 percent), and exam preparation (76 percent). Research and revising see adoption rates around 70 percent. At the other extreme, only 23 percent report using AI during office hours, and 32 percent use it for campus resources like tutoring centers and library services. This pattern suggests that students adopt AI primarily for independent work rather than for tasks involving direct interaction with instructors or peers. Intensity of use follows a similar pattern: among AI users, average intensity is highest for writing (41 out of 100) and problem sets (34), and lowest for office hours (11) and lectures (13).

Students are more likely to use AI for time-intensive tasks that they perceive as important. Figure 2, Panel A shows a positive relationship between time spent on a task and AI adoption: tasks that consume more hours per week see higher rates of AI use. Panels B–E show that AI adoption increases with perceived task importance, though the relationship is strongest for grades (Panel D) and the overall importance index (Panel B), and weaker for learning (Panel C) and skills (Panel E). The relationship with enjoyment is essentially flat (Panel F). These patterns suggest that students deploy AI strategically on consequential, time-consuming tasks rather than using it indiscriminately.

5.2 Effects of AI on Time and Cognitive Effort

We next examine how AI affects the time students spend on academic tasks and the cognitive effort they exert. These estimates correspond to the time compression parameter τ_i and the effort substitution parameter μ_i in our model. Figure 3 presents the results separately for each task. Panel A displays mean cognitive effort on a zero-to-ten scale, with red markers showing effort without AI (counterfactual) and blue markers showing effort with AI (actual). Panel B displays the analogous comparison for time spent in hours per week.

AI reduces both time and effort across nearly all tasks. On average, students report exerting 6.8 units of cognitive effort with AI compared to 7.7 units without—a reduction of 0.9 points on the ten-point scale, or roughly 11 percent. Time savings are proportionally larger: students report spending 5.7 hours per week on tasks with AI compared to 6.8 hours without, a reduction of 1.1 hours or 16 percent. These averages mask substantial heterogeneity across tasks. The largest effort reductions occur for office hours, recordings, and revising written work, while effort reductions are smaller for labs, exam preparation, and lectures. Time savings are largest for problem sets, research, and revising, and smallest for lectures and notes.[2]

A natural question is whether time savings and effort reductions are independent margins or whether they move together. Figure 4 plots the relationship between changes in time and changes in effort across task observations. Tasks on which AI saves more time also show larger reductions in cognitive effort: a one-hour reduction in time spent is associated with roughly a 0.10 point reduction in effort on the ten-point scale. This positive correlation suggests that time savings and effort reductions may be two sides of the same coin—AI saves time precisely by reducing the cognitive work required to complete a task. If effort is an input to learning, this finding implies that the benefits of AI (time savings) may inevitably come bundled with costs (reduced cognitive engagement).

Heterogeneity by Task Characteristics. These average reductions may mask heterogeneity across task types. Figure 5 plots the mean change in cognitive effort and time separately for tasks above and below the median of each characteristic. AI reduces cognitive effort across all task types, with point estimates ranging from -0.72 to -1.03 on

[2] Appendix Figures A4 and A5 display the full distributions of effort and time changes. The distributions are heavily left-skewed: the vast majority of task observations show reductions in both effort and time, with only a small share showing increases.

the ten-point scale. The reductions are consistently larger for tasks rated below median importance—roughly 0.2 to 0.3 points larger across most dimensions (with the exception of skills importance). Time savings show a different pattern. Students save substantially more time on tasks they already spend more time on (1.3 hours above median versus 0.9 hours below), but the relationship with perceived importance is weaker and inconsistent across dimensions. This asymmetry—importance-dependent effort reductions but time savings that scale mainly with baseline time use—suggests that AI reduces cognitive engagement more on tasks students view as less consequential, while time savings are driven primarily by how much time a task demands rather than how important students perceive it to be.

Heterogeneity by Student Characteristics. Heterogeneity by student characteristics reveals a different pattern. Figure 6 plots the mean change in cognitive effort and time by demographic and academic characteristics. AI reduces cognitive effort by roughly 0.8 to 1.0 points for all student groups: male and female, white and non-white, first-generation and continuing-generation, above- and below-median SAT and GPA, and STEM and non-STEM majors. None of the pairwise differences is large relative to the overall mean reduction of 0.9 points. Time savings vary more substantially. First-generation students report saving 1.5 hours per week compared to 1.0 hours for continuing-generation students. The most pronounced differences are by academic preparation and field of study: students with above-median SAT scores save 1.6 hours compared to 0.6 hours for those below the median, and STEM majors save 1.6 hours compared to 0.8 hours for non-STEM majors.

To examine how student and task characteristics jointly relate to changes in time and effort, in Table 3 we estimate OLS regressions that include all observed characteristics simultaneously. The dependent variables are the change in mental effort (columns 1–2) and the change in time spent (columns 3–4), calculated as current values minus counterfactual values without AI. Odd-numbered columns include student and task characteristics; even-numbered columns add academic task fixed effects.

The regression estimates confirm that AI substantially reduces both effort and time. Conditional on all covariates, using AI for a task is associated with a 0.7-point reduction in mental effort and a 0.76–0.86 hour reduction in weekly time spent. Students who report using AI more than they would like experience even larger time savings—an additional 0.9 hours per week—though this variable does not predict effort reductions. This pattern suggests that students who recognize they are over-relying on AI are precisely those experiencing the largest behavioral changes. Task characteristics explain little variation in effort

changes, but matter for time: students save more time on tasks they rate as important for skill development and less time on tasks they find enjoyable. Among student characteristics, male students report significantly larger effort reductions than female students (0.14 points). The most striking result concerns field of study: STEM majors report time savings roughly 0.3 hours smaller than non-STEM majors, even conditional on task fixed effects and other characteristics.

5.3 Cognitive Effort by Type of Thinking

The previous section documented that AI reduces cognitive effort on average, but this average may mask important heterogeneity across different types of thinking. If AI primarily substitutes for lower-order cognitive activities like remembering and understanding while preserving engagement in higher-order activities like analyzing and evaluating, then the reduction in overall effort may be less concerning for human capital formation. We examine this possibility using six cognitive categories based on Bloom's taxonomy. For each task on which they use AI, students rated whether AI requires more or less mental effort compared to completing the task without AI, separately for each cognitive category. Responses range from one (much less effort with AI) to five (much more effort with AI). Figure 7 presents the distribution of responses.

The pattern does not follow a simple hierarchy from lower- to higher-order thinking. For remembering—the lowest level of Bloom's taxonomy—29 percent of task observations report reduced effort with AI. The share reporting reduced effort is highest for understanding and synthesizing (both 37 percent), with applying and analyzing at intermediate rates (35 and 34 percent, respectively). Evaluating shows a distinct pattern: only 24 percent report reduced effort, while 38 percent report increased effort—the highest rate of increased effort across all categories. This suggests that AI most readily substitutes for tasks involving comprehension and integration of material, but less so for tasks requiring critical judgment.

To identify which students and tasks exhibit effort reductions across cognitive levels, in Table 4 we estimate OLS regressions predicting whether AI reduced effort for each Bloom's taxonomy category. The dependent variable in each column is an indicator equal to one if the student reported reduced effort with AI for that cognitive activity. All models include the full set of student, task, and academic controls.

Using AI for a task significantly predicts effort reductions across all cognitive categories, but the magnitude varies. The coefficient is largest for synthesizing (0.37), followed

closely by understanding (0.36). The coefficients for solving (0.34) and analyzing (0.33) are also substantial, while the smallest effects occur at the extremes of the taxonomy: evaluating (0.24) and remembering (0.28). Notably, students who report using AI more than they would like do not exhibit larger effort reductions on any cognitive activity, suggesting that perceived overuse does not translate into differential disengagement across types of thinking. Task and student characteristics explain little additional variation; none of the importance measures, demographic variables, or academic characteristics significantly predicts effort reductions across cognitive levels.

5.4 Reallocation of Time Savings

The model in Section 2 shows that AI's aggregate effect on learning depends not only on how it changes time and effort within activities, but also on how students reallocate time freed by AI. If students reinvest time savings into other learning activities, aggregate learning may increase even if AI reduces effort on individual tasks. If instead students reallocate time to leisure, the aggregate effect depends on whether the marginal return to rest exceeds the marginal return to additional study. The reallocation parameter η—the share of freed time allocated to academics—is therefore central to understanding AI's net effect on human capital formation.

We measure reallocation by asking students who report that AI saves them time to indicate what percentage of saved time they allocate to each of nine categories. The vast majority of AI users (83 percent) report that AI saves them time, with average savings of approximately five hours per week. Figure 8 shows where this freed time goes. Leisure absorbs the largest share, accounting for 30 percent of reallocated time or roughly 64 minutes per week. Sleep is second at 20 percent (44 minutes), followed by social activities at 17 percent (37 minutes), personal care at 13 percent (29 minutes), and paid work at 13 percent (28 minutes). Extracurricular activities account for 8 percent. Strikingly, other academic activities receive only 7 percent of the time savings—roughly 14 minutes per week. This implies a reallocation parameter of $\hat{\eta} \approx 0.07$, meaning that over 90 percent of time freed by AI flows to non-academic pursuits.

This finding has important implications for interpreting AI's effect on learning. Even if AI increases the productivity of effort within tasks (the π_i effect in the model), the near-complete reallocation of time savings away from academics means that students are not using AI to study more. Whether this reallocation harms aggregate learning depends on the counterfactual productivity of the reallocated time. If students were severely sleep-

deprived at baseline, for instance, the reallocation to sleep we document could improve the productivity of remaining study hours enough to offset the reduction in study time—particularly if the study time displaced by AI was low-productivity to begin with. But leisure and social activities absorb the largest share of freed time, and it is difficult to construct a plausible channel through which additional leisure enhances academic human capital. Given these magnitudes, we view it as unlikely that AI-induced productivity gains within tasks fully offset the reallocation of time away from learning activities.

5.5 Perceived Value and Self-Assessed Overuse

The time and effort savings documented above raise a natural question: how much do students value AI tools? We asked students to estimate the monetary value they receive from generative AI in a typical month, taking into account time saved, quality of work produced, and any other benefits.

Students report substantial perceived value from AI tools (Appendix Figure A7, Panel A). The winsorized mean estimated value is $80 per month, and the median is $25. The distribution is heavily right-skewed, with roughly one-quarter of students reporting zero value and a long tail exceeding $500 per month. These valuations are lower than a simple calculation would imply: if AI saves five hours per week and students value their time at $25 per hour, the implied monthly value would be roughly $500. The gap may reflect that students do not fully monetize time savings—particularly if freed time flows to leisure rather than paid work—or that they perceive costs from AI use that offset the benefits. Perceived value varies substantially across student characteristics (Appendix Figure A7, Panel B). Male students report higher valuations than female students ($93 vs. $72), and students with below-median GPAs report values of $91 compared to $69 for those above the median. First-generation students and those from public high schools also report higher valuations.

Despite these high valuations, a substantial fraction of students report using AI more than they believe they should (Appendix Figure A8). While 58.6 percent report using AI the right amount, 254.6 percent report using it "somewhat too much" and an additional 6 percent say "far too much." Only 10.8 percent report wanting to use AI more than they currently do. This pattern suggests a self-control problem: students recognize that their AI use may be excessive for their learning goals, yet they continue to use it because the immediate benefits of time savings outweigh the diffuse costs of reduced skill development.

The combination of high perceived value and self-assessed overuse presents a puzzle.

If students benefit so much from AI, why do many also believe they use it too much? One possibility is that students distinguish between short-run and long-run benefits. AI provides immediate gratification through reduced effort and faster task completion, but students may recognize that this comes at the cost of learning that would benefit them in the future. This interpretation is consistent with the finding in Table 3 that students who report using AI too much experience significantly larger time savings but not larger effort reductions—they are aware that AI is changing their behavior in ways they do not fully endorse.

6 Conclusion

This paper documents how college students change their time allocation and cognitive effort when using generative AI for academic work. We find that students deploy AI strategically, adopting it primarily for time-intensive, independent tasks they view as consequential—readings, problem sets, writing, exam preparation—rather than for tasks involving direct interaction with instructors or peers. AI reduces both time and effort substantially and these reductions are positively correlated, suggesting that AI saves time precisely by reducing the cognitive work required to complete a task. Students reallocate nearly all time saved by AI to non-academic activities such as leisure, sleep, and social activities. Despite reporting high valuations of AI tools, many students believe they use AI more than they should. This self-awareness suggests that interventions helping students calibrate their AI use—rather than banning AI outright—may be more effective.

Our analysis has limitations. Our measures rely on self-reports, which may be subject to recall bias. Our sample may not represent the broader population of college students. We measure behavior at a single point in time; how students integrate AI into their academic lives may evolve as the technology improves and norms develop. Despite these limitations, our findings highlight that AI's effects on human capital formation depend not only on the technology's capabilities but also on how students choose to use it—and crucially, on what they do with the time it frees.

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Silvana Guadalupe Navarro Jiménez, Irena Nikaj, Maria Mihaylova Nisheva, Efi Nisiforou, Joseph Njiku, Singhanat Nomnian, Lulzime Nuredini-Mehmedi, Ernest Nyamekye, Alka Obadić, Abdelmohsen Hamed Okela, Dorit Olenik-Shemesh, Izabela Ostoj, Kevin Javier Peralta-Rizzo, Almir Peštek, Amila Pilav-Velić, Dilma Rosanda Miranda Pires, Eyal Rabin, Daniela Raccanello, Agustine Ramie, Md. Mamun Ur Rashid, Robert A. P. Reuter, Valentina Reyes, Ana Sofia Rodrigues, Paul Rodway, Silvia Ručinská, Shorena Sadzaglishvili, Ashraf Atta M. S. Salem, Gordana Savić, Astrid Schepman, Samia Mokhtar Shahpo, Abdelmajid Snouber, Emma Soler, Bengi Sonyel, Eliza Stefanova, Anna Stone, Artur Strzelecki, Tetsuji Tanaka, Carolina Tapia Cortes, Andrea Teira-Fachado, Henri Tilga, Jelena Titko, Maryna Tolmach, Dedi Turmudi, Laura Varela-Candamio, Ioanna Vekiri, Giada Vicentini, Erisher Woyo, Özlem Yorulmaz, Said A. S. Yunus, Ana-Maria Zamfir, Munyaradzi Zhou, and Aleksander Aristovnik, "Higher Education Students' Perceptions of ChatGPT: A Global Study of Early Reactions," *PLOS ONE*, February 2025, *20* (2), e0315011.

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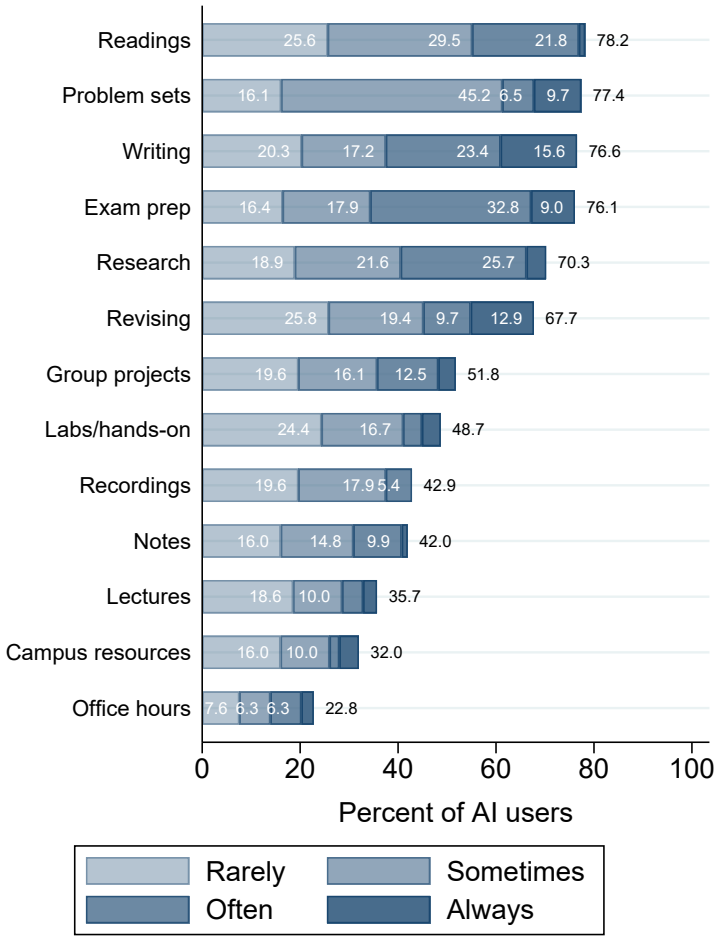
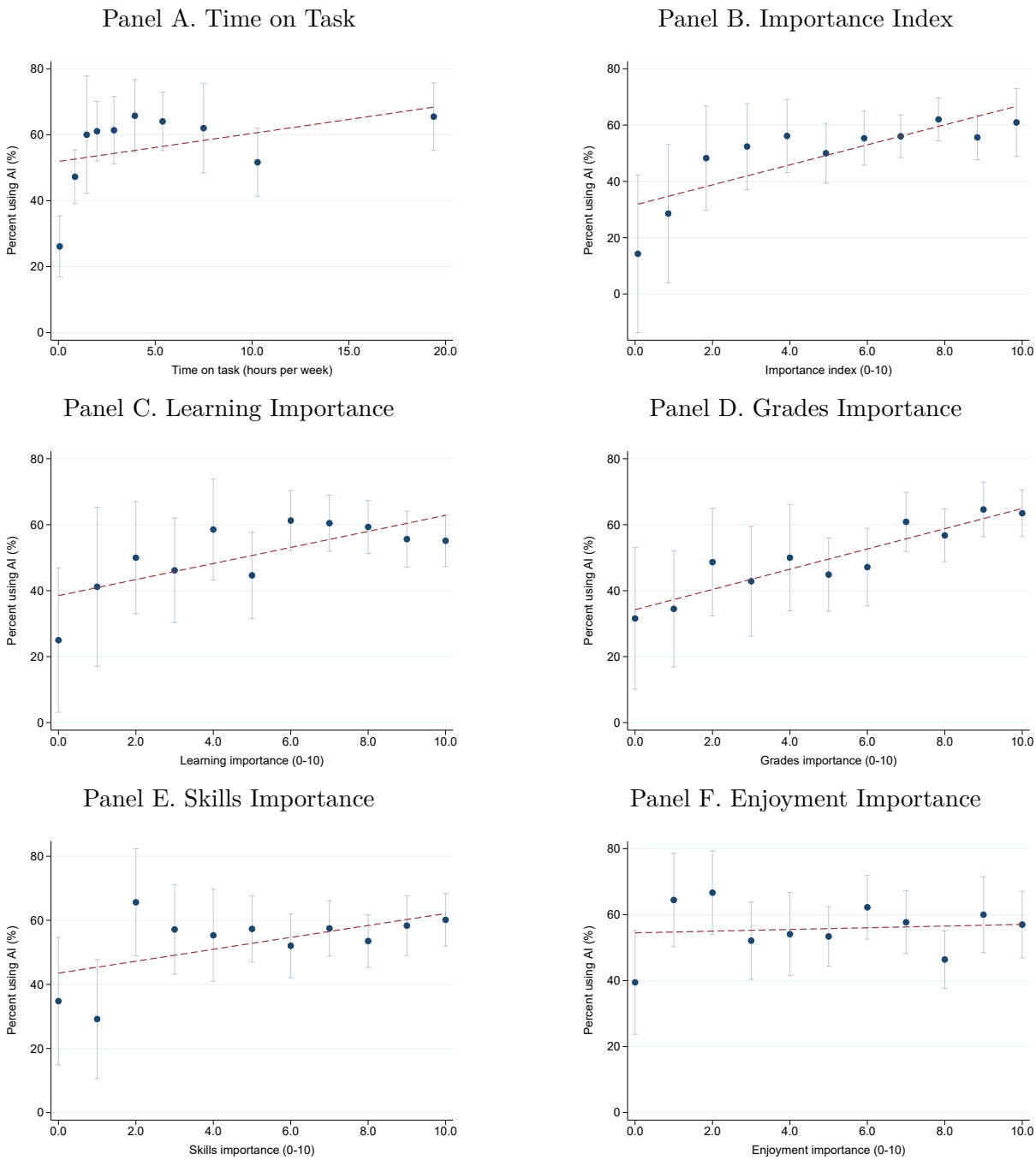
Figures and Tables

Figure 1: AI Use Frequency Across Academic Tasks



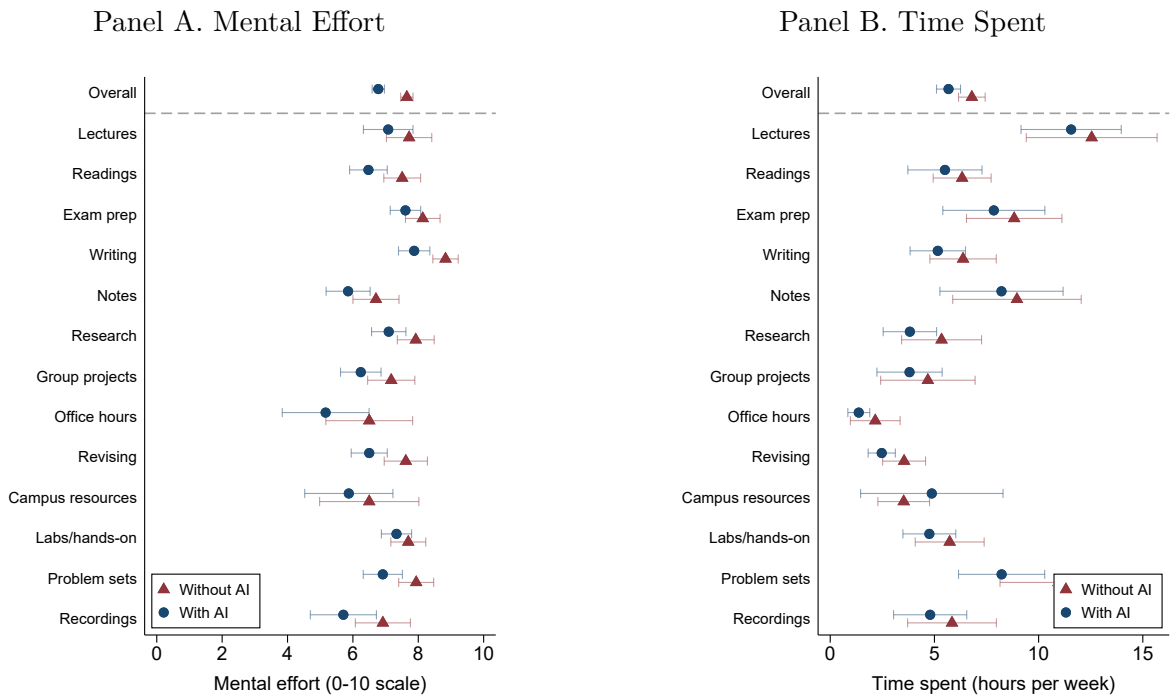
Notes: This figure displays the distribution of AI use frequency for each academic task. Response categories correspond to the share of times AI is used: Rarely (1–2 times out of 10), Sometimes (3–5 times), Often (6–8 times), and Always (9–10 times). Tasks are sorted by total AI adoption rate in descending order. Tasks include: (1) Lectures: attend lectures, seminars, and laboratory sessions; (2) Readings: complete assigned readings from textbooks and academic journals; (3) Exam prep: prepare for exams and assessments by reviewing material and creating study aids; (4) Writing: write research papers, essays, and reports; (5) Notes: take notes during lectures and review sessions; (6) Research: search for, evaluate, and synthesize academic sources; (7) Group projects: collaborate with peers on team assignments; (8) Office hours: meet with professors or TAs to seek clarification; (9) Revising: revise and edit written work based on feedback; (10) Campus resources: utilize tutoring centers, writing labs, and academic support services; (11) Labs/hands-on: complete laboratory experiments, field work, or studio projects; (12) Problem sets: solve homework exercises and practice problems; (13) Recordings: watch recorded lectures and instructional videos. Sample includes all task observations with non-missing time data.

Figure 2: AI Adoption and Task Characteristics



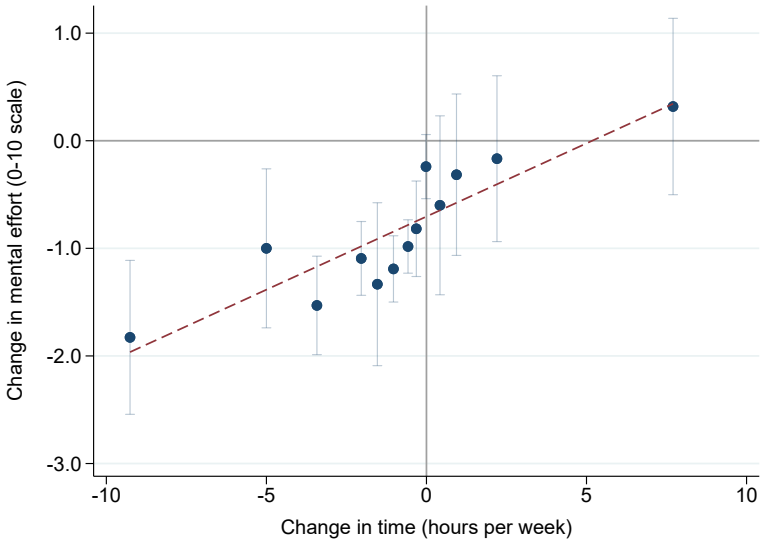
Notes: Each panel shows a binned scatterplot of AI adoption against task characteristics. In Panel A, observations are grouped into ten equal-sized bins based on time spent; in Panels B–F, observations are grouped by the integer value of each importance measure. Points represent bin means; vertical lines show 95 percent confidence intervals. The dashed line shows a linear fit. Panel A plots AI adoption against time spent on the task. Panel B plots AI adoption against the mean importance index, which averages ratings across all four importance dimensions. Panels C–F plot AI adoption against each importance dimension separately.

Figure 3: How AI Affects Time and Effort Across Academic Tasks



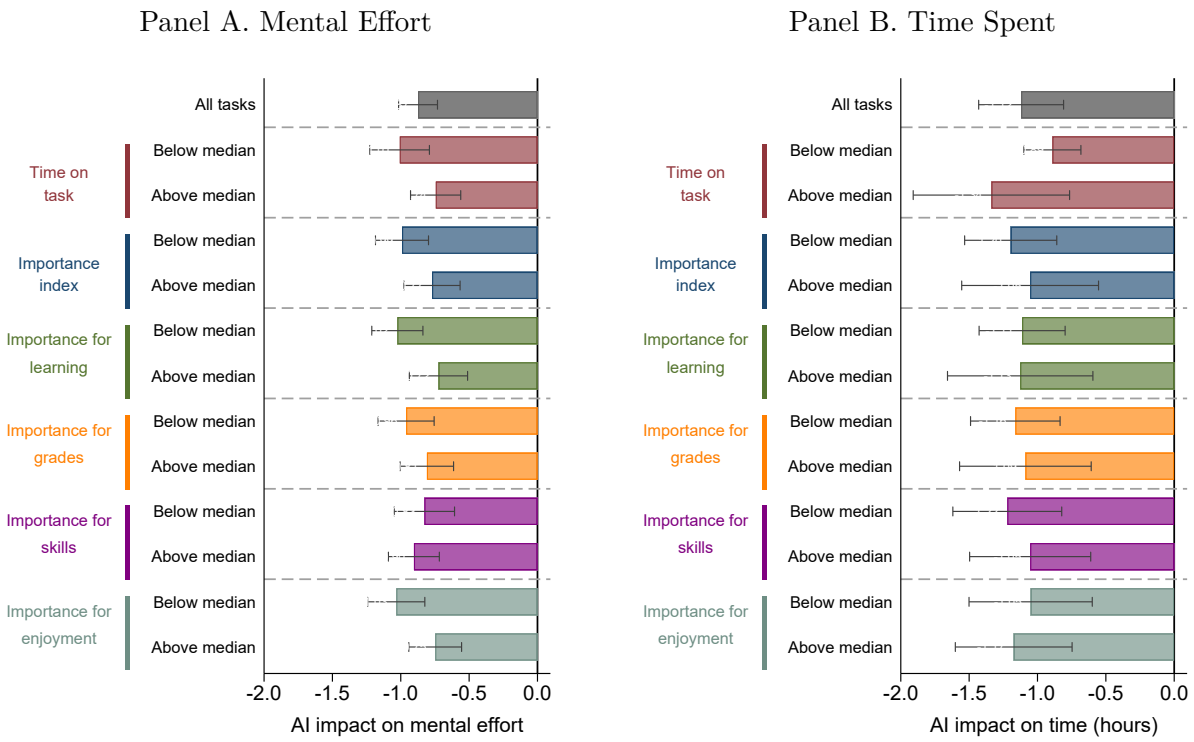
Notes: This figure shows self-reported mental effort and time spent on academic tasks with and without AI. The sample includes only students who use AI for each task. Maroon points show mean outcomes without AI (counterfactual); navy points show mean outcomes with AI (actual). Horizontal lines connect the two conditions for each task. Error bars represent 95 percent confidence intervals. The dashed horizontal line separates the overall average from task-specific estimates.

Figure 4: Relationship Between Time Savings and Effort Reduction



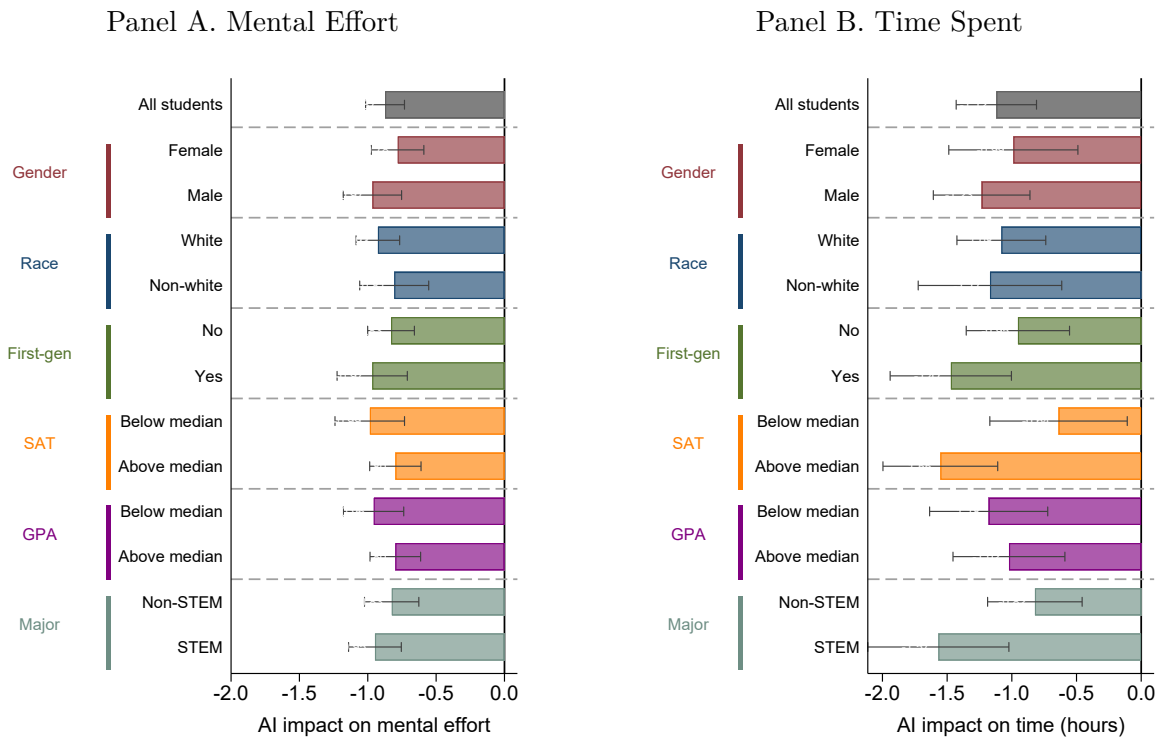
Notes: This figure shows a binned scatterplot of the change in mental effort against the change in time spent due to AI. Observations are grouped into equal-sized bins based on time change; points represent bin means. Vertical lines show 95 percent confidence intervals. The dashed line shows a linear fit. Negative values on both axes indicate reductions with AI relative to without AI. The sample includes only AI users with non-missing data on both effort and time.

Figure 5: AI Impact by Task Characteristics



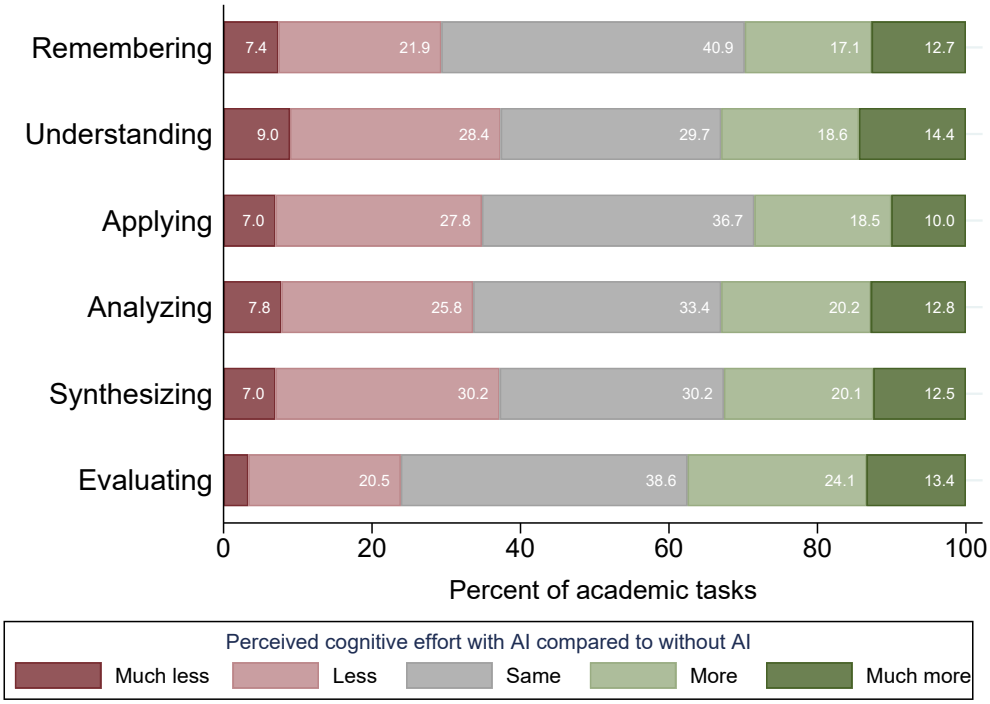
Notes: This figure shows the mean change in self-reported mental effort (Panel A) and time spent (Panel B) due to AI, by task characteristics. Negative values indicate AI reduces effort or time. The sample includes only AI users with non-missing data. Each category splits tasks by median value: time spent on task, overall importance index, and importance for learning, grades, skills, and enjoyment. Error bars represent 95 percent confidence intervals. The vertical line at zero indicates no change.

Figure 6: AI Impact by Student Characteristics



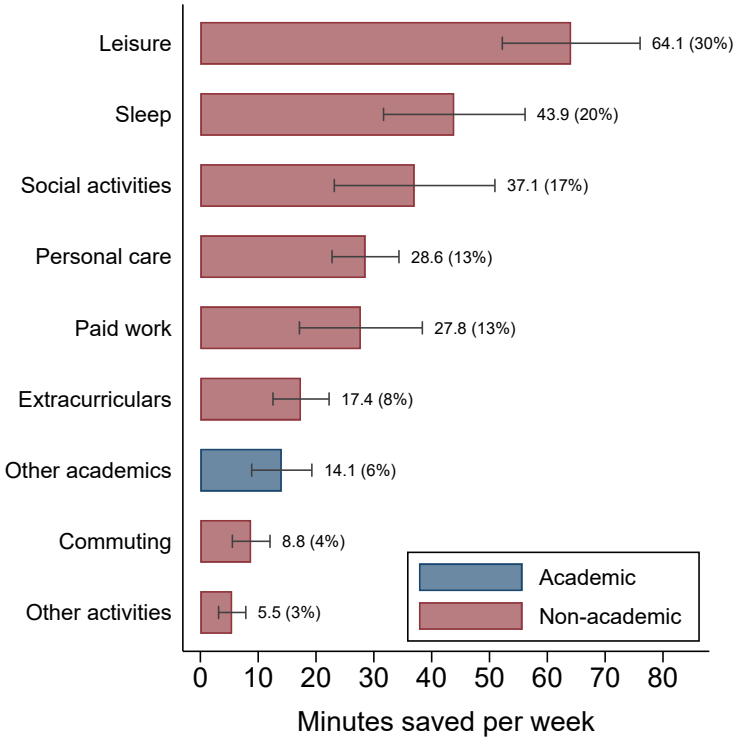
Notes: This figure shows the mean change in self-reported mental effort (Panel A) and time spent (Panel B) due to AI, by student characteristics. Negative values indicate AI reduces effort or time. The sample includes only AI users with non-missing data. SAT and GPA split students at their respective medians. "First-gen" indicates first-generation college student status. Error bars represent 95 percent confidence intervals. The vertical line at zero indicates no change.

Figure 7: Perceived Cognitive Effort When Using Generative AI



Notes: This figure shows the distribution of students' perceived cognitive effort when using generative AI compared to working without AI, across six cognitive activities based on Bloom's taxonomy. For each activity, students rated their effort on a 5-point scale: 1 = "Much less effort with AI," 2 = "Less effort," 3 = "Same effort," 4 = "More effort," 5 = "Much more effort." Activities are ordered from lower-order thinking skills (Remembering) to higher-order skills (Evaluating). Sample includes all task observations where the student reported using AI.

Figure 8: Reallocation of Time Saved Due to AI



Notes: This figure shows how students report reallocating time saved by using AI. The sample includes only AI users. Respondents indicated what percentage of saved time they allocate to each category. Navy bars represent academic uses; maroon bars represent non-academic uses. Categories are sorted by mean share in descending order.

Table 1: Summary Statistics by AI Usage

	All (1)	AI Users (2)	Non-Users (3)
Panel A. Demographics			
Age	21.2	21.1	21.6
Male	0.449	0.475	0.273
Female	0.527	0.518	0.576
White	0.584	0.576	0.667
Black	0.182	0.187	0.152
Hispanic/Latino	0.122	0.117	0.152
International student	0.003	0.004	0.000
Public high school	0.888	0.875	0.969
Private high school	0.085	0.098	0.000
First-generation student	0.345	0.339	0.424
Panel B. Academic Characteristics			
GPA (4.0 scale)	3.470	3.466	3.544
SAT score (converted)	1251.4	1257.5	1207.6
Hours on academics per week	26.2	26.3	26.3
STEM major	0.391	0.405	0.303
Social science major	0.320	0.346	0.152
Humanities/Arts major	0.108	0.078	0.364
Panel C. AI Usage			
Heavy AI user	0.531	0.599	0.000
AI saves time	0.829	0.829	0.000
Currently pays for AI	0.179	0.202	0.000
Monthly AI spending (USD)	3.2	3.6	0.0
Panel D. AI Models Used			
Uses ChatGPT (free)	0.734	0.829	0.000
Uses ChatGPT (paid)	0.183	0.206	0.000
Uses Claude (free)	0.110	0.125	0.000
Uses Gemini (free)	0.341	0.385	0.000
Uses NotebookLM	0.072	0.082	0.000
Uses any paid AI	0.239	0.276	0.000
N (students)	297	257	33

Notes: This table presents summary statistics for the survey sample by AI usage status. Column 1 reports means for all students. Column 2 reports means for students who use AI for academic purposes during the term. Column 3 reports means for students who never use AI for academics. Panel A reports demographic characteristics. Panel B provides academic characteristics, including GPA on a 4.0 scale, SAT scores (with ACT converted to SAT equivalents), and weekly hours spent on academics. Panel C summarizes AI usage intensity among users. Panel D shows the proportion of students using specific AI models. For binary variables, the mean represents the proportion of the sample.

Table 2: Descriptive Statistics by Academic Task

			AI Usage		Task Importance			
	Time (hrs/wk) (1)	Effort (0–10) (2)	Uses AI (%) (3)	Intensity (0–100) (4)	Learning (0–10) (5)	Grades (0–10) (6)	Skills (0–10) (7)	Enjoyment (0–10) (8)
Lectures	11.14	6.49	35.7	12.50	8.11	7.66	7.03	7.10
Readings	5.13	6.38	78.2	32.12	6.78	6.82	6.49	5.41
Exam prep	7.35	7.52	76.1	41.12	7.51	8.28	6.42	5.81
Writing	5.29	8.00	76.6	41.17	6.80	7.58	6.64	5.27
Notes	6.88	5.68	42.0	16.42	7.58	7.40	6.38	5.99
Research	3.43	7.01	70.3	33.31	6.95	6.81	7.46	5.82
Group projects	3.29	6.34	51.8	21.52	5.91	6.32	7.34	5.70
Office hours	1.15	4.81	22.8	10.51	6.24	5.67	6.28	5.76
Revising	2.42	6.47	67.7	30.65	6.85	6.97	7.26	5.42
Campus resources	2.46	4.86	32.0	11.60	5.24	5.44	5.08	4.72
Labs/hands-on	3.99	6.95	48.7	16.67	7.51	6.55	7.53	7.05
Problem sets	8.14	7.15	77.4	34.19	7.68	7.69	6.37	5.15
Recordings	5.47	5.39	42.9	13.84	7.07	7.21	5.68	5.36
Overall	5.12	6.40	55.5	24.26	6.99	6.97	6.65	5.79

Notes: This table reports mean values for each academic task. Time is hours spent per week on the task. Effort is self-reported mental effort on a 0–10 scale. "Uses AI" is the percentage of students who report using AI for the task. "Intensity" is the average frequency of AI use (out of 100 time doing the task), with non-users coded as zero. Task importance measures are on a 0–10 scale, where higher values indicate greater importance for learning, course grades, developing skills, or personal enjoyment. See Appendix Table A1 for a detail description of each activity.

Table 3: Predictors of Change in Time and Effort Due to AI

	Δ Mental Effort		Δ Time Spent	
	(1)	(2)	(3)	(4)
Panel A. AI usage				
Uses AI for task	-0.712***	-0.740***	-0.857***	-0.756***
	(0.096)	(0.102)	(0.221)	(0.231)
Uses too much AI	-0.495**	-0.528**	-0.953**	-0.856*
	(0.212)	(0.217)	(0.431)	(0.445)
Panel B. Task characteristics				
Learning importance	0.072	0.091	-0.028	0.084
	(0.081)	(0.087)	(0.165)	(0.170)
Grades importance	0.021	0.035	-0.039	-0.033
	(0.075)	(0.077)	(0.113)	(0.111)
Skills importance	-0.097	-0.125	0.120	0.155
	(0.080)	(0.087)	(0.135)	(0.148)
Enjoyment importance	0.110	0.099	-0.051	-0.161
	(0.091)	(0.100)	(0.183)	(0.186)
Panel C. Student characteristics				
Age	0.031	0.030	-0.016	-0.007
	(0.035)	(0.035)	(0.049)	(0.051)
Male	-0.148	-0.143	0.082	0.089
	(0.096)	(0.094)	(0.260)	(0.253)
White	-0.087	-0.107	-0.071	-0.061
	(0.104)	(0.104)	(0.262)	(0.268)
First-generation	-0.014	-0.011	-0.323	-0.297
	(0.106)	(0.107)	(0.246)	(0.245)
Patience	-0.092	-0.094	0.100	0.073
	(0.056)	(0.058)	(0.141)	(0.138)
Motivation to learn	-0.006	-0.011	-0.014	-0.040
	(0.071)	(0.073)	(0.163)	(0.161)
Panel D. Academic characteristics				
GPA	0.088	0.092	0.094	0.082
	(0.064)	(0.063)	(0.100)	(0.097)
SAT score	0.082	0.079	-0.214*	-0.217*
	(0.054)	(0.055)	(0.121)	(0.115)
STEM major	-0.108	-0.101	-0.324	-0.311
	(0.095)	(0.099)	(0.259)	(0.255)
Mean dep. var.	-0.49	-0.49	-0.61	-0.61
Task fixed effects	No	Yes	No	Yes
N	745	745	747	747
R-squared	0.164	0.175	0.068	0.091

Notes: This table reports OLS estimates of the predictors of changes in mental effort and time spent due to AI. The dependent variables are the change in mental effort (columns 1–2) and the change in time spent in hours (columns 3–4), calculated as current values minus counterfactual values without AI; negative values indicate that AI reduces the outcome. Uses too much AI" is a dummy equal to one if the student self-reports using AI far too much" or "somewhat too much" (zero for non-AI users). All continuous variables are standardized to have mean zero and standard deviation one. Standard errors clustered at the student level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Reduced Effort by Bloom's Taxonomy Level

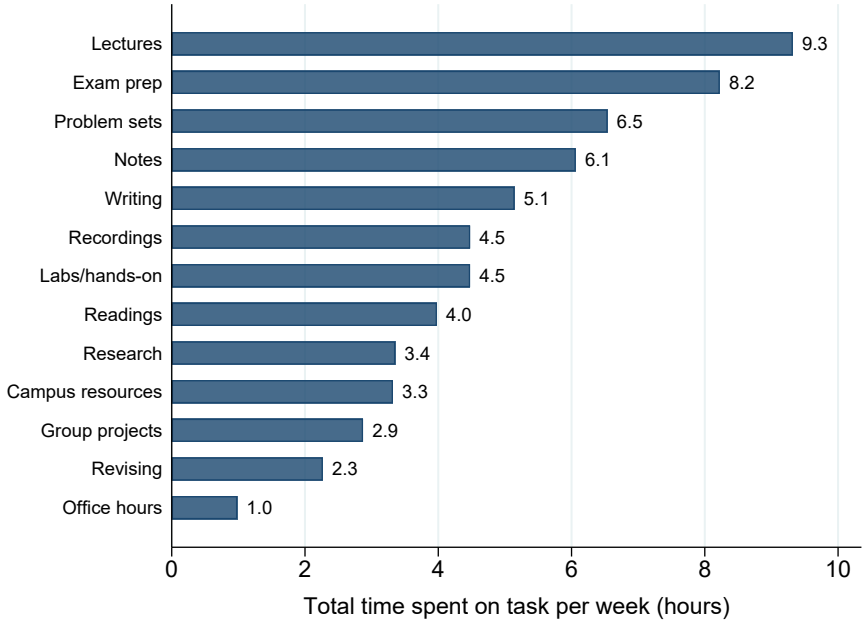
	Outcome: = 1 if AI reduces mental effort on...					
	Remember (1)	Understand (2)	Solve (3)	Analyze (4)	Evaluate (5)	Synthesize (6)
Panel A. AI usage						
Uses AI for task	0.279***	0.358***	0.341***	0.333***	0.236***	0.373***
	(0.033)	(0.036)	(0.037)	(0.033)	(0.029)	(0.037)
Uses too much AI	0.112**	0.046	0.070	0.070	0.004	0.020
	(0.054)	(0.057)	(0.053)	(0.050)	(0.049)	(0.057)
Panel B. Task characteristics						
Learning importance	0.008	0.024	-0.001	-0.012	-0.010	-0.005
	(0.024)	(0.026)	(0.026)	(0.026)	(0.020)	(0.025)
Grades importance	-0.027	-0.009	0.000	-0.002	-0.010	0.002
	(0.018)	(0.020)	(0.018)	(0.020)	(0.017)	(0.017)
Skills importance	-0.010	0.009	0.029	0.020	0.034*	-0.002
	(0.018)	(0.019)	(0.020)	(0.019)	(0.018)	(0.020)
Enjoyment importance	0.013	-0.031	-0.013	-0.025	-0.009	-0.016
	(0.021)	(0.022)	(0.023)	(0.022)	(0.018)	(0.024)
Panel C. Student characteristics						
Age	0.011	-0.008	-0.005	-0.008	-0.006	-0.016*
	(0.007)	(0.008)	(0.008)	(0.008)	(0.006)	(0.009)
Male	0.023	0.022	0.025	-0.024	-0.000	0.015
	(0.031)	(0.034)	(0.035)	(0.031)	(0.026)	(0.036)
White	0.027	0.011	0.049	0.050	0.012	0.041
	(0.029)	(0.034)	(0.036)	(0.033)	(0.026)	(0.034)
First-generation	0.005	0.021	-0.004	0.020	0.043	0.028
	(0.033)	(0.033)	(0.034)	(0.034)	(0.027)	(0.035)
Patience	0.002	-0.033**	-0.008	-0.021	-0.022*	-0.006
	(0.015)	(0.017)	(0.016)	(0.016)	(0.013)	(0.019)
Motivation to learn	-0.001	-0.012	-0.020	-0.008	-0.015	0.004
	(0.017)	(0.017)	(0.016)	(0.016)	(0.015)	(0.018)
Panel D. Academic characteristics						
GPA	0.004	0.003	-0.006	0.000	-0.009	-0.002
	(0.015)	(0.014)	(0.015)	(0.013)	(0.012)	(0.015)
SAT score	0.021	0.014	0.019	0.024	0.008	0.003
	(0.015)	(0.016)	(0.016)	(0.016)	(0.012)	(0.018)
STEM major	-0.017	-0.002	-0.008	0.010	-0.018	0.013
	(0.030)	(0.034)	(0.035)	(0.033)	(0.025)	(0.034)
Mean dep. var.	0.16	0.21	0.19	0.19	0.13	0.21
N	705	729	699	731	719	729
R-squared	0.196	0.239	0.237	0.224	0.148	0.233

Notes: This table reports OLS estimates of the predictors of reduced cognitive effort when using AI. The dependent variable in each column is a dummy equal to one if the student reports exerting less effort with AI ("much less" or "somewhat less") relative to without, and zero if the same or more effort. Columns correspond to Bloom's taxonomy categories: remembering facts, understanding concepts, solving problems, analyzing information, evaluating arguments, and synthesizing ideas. The sample includes only AI users. "Uses too much AI" is a dummy equal to one if the student self-reports using AI "far too much" or "somewhat too much." All continuous variables are standardized to have mean zero and standard deviation one. Standard errors clustered at the student level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

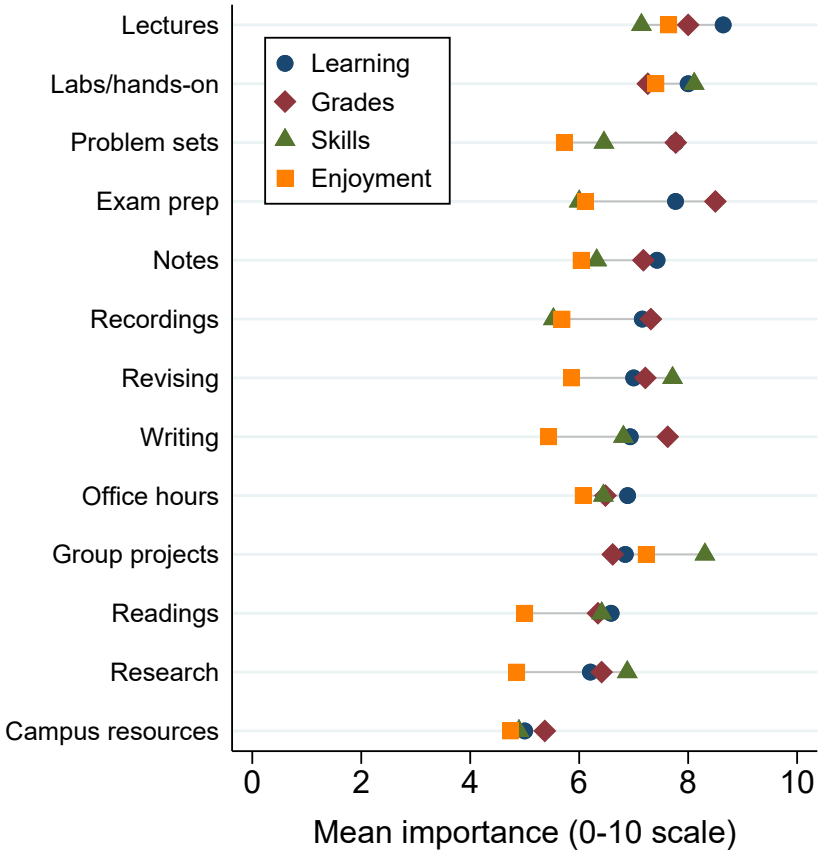
A Appendix Figures and Tables

Figure A1: Distribution of Time Across Academic Tasks



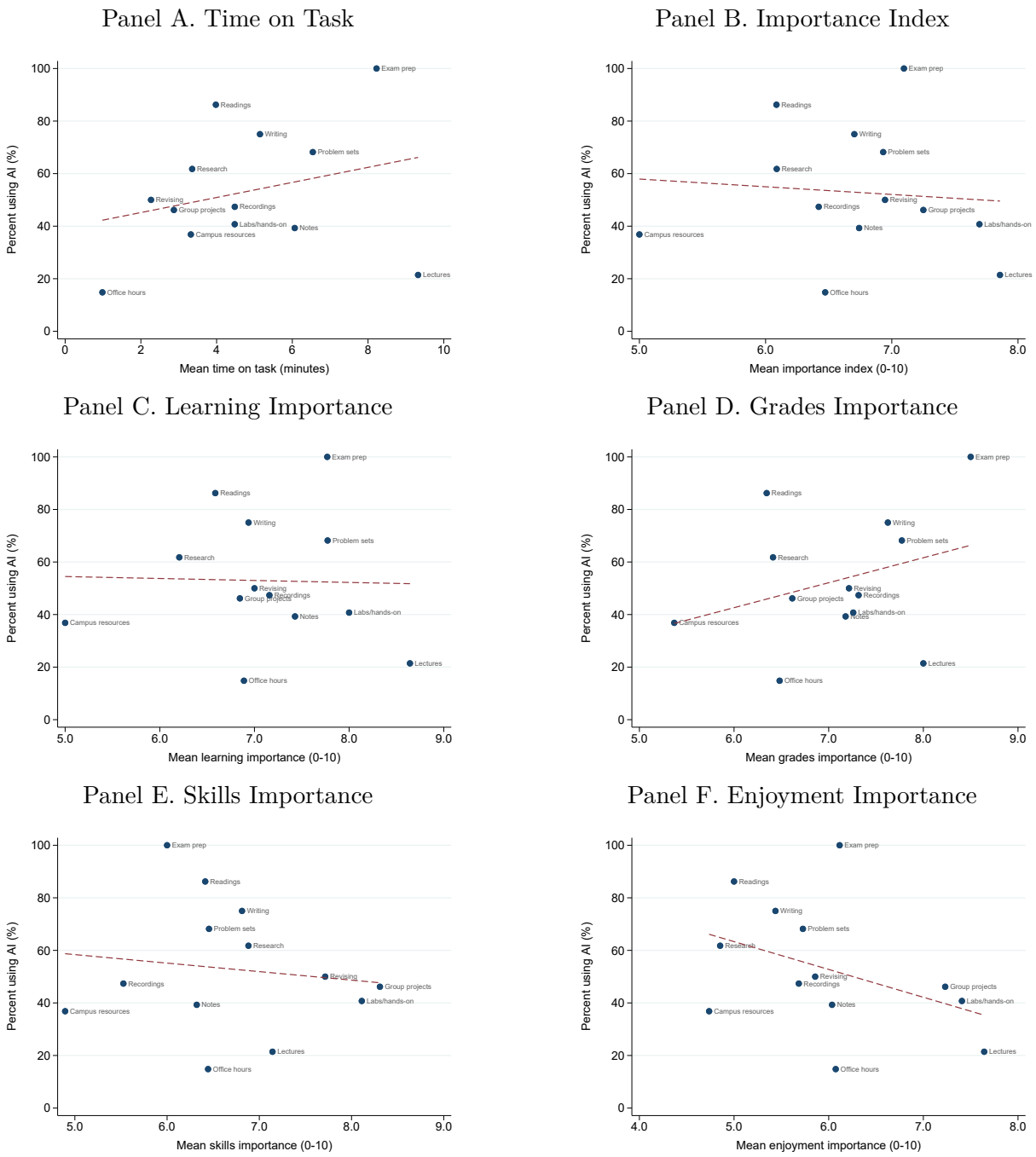
Notes: This figure shows the average weekly time students report spending on each academic task. Tasks include: (1) Lectures: attend lectures, seminars, and laboratory sessions; (2) Readings: complete assigned readings from textbooks and academic journals; (3) Exam prep: prepare for exams and assessments by reviewing material and creating study aids; (4) Writing: write research papers, essays, and reports; (5) Notes: take notes during lectures and review sessions; (6) Research: search for, evaluate, and synthesize academic sources; (7) Group projects: collaborate with peers on team assignments; (8) Office hours: meet with professors or TAs to seek clarification; (9) Revising: revise and edit written work based on feedback; (10) Campus resources: utilize tutoring centers, writing labs, and academic support services; (11) Labs/hands-on: complete laboratory experiments, field work, or studio projects; (12) Problem sets: solve homework exercises and practice problems; (13) Recordings: watch recorded lectures and instructional videos. Sample includes all task observations with non-missing time data.

Figure A2: Perceived Importance of Academic Tasks



Notes: This figure shows the mean perceived importance of each academic task across four dimensions: learning, grades, skills, and enjoyment. Students rated each dimension on a 0–10 scale, where 0 = "Not at all important" and 10 = "Extremely important." Tasks are ordered by importance for learning. Horizontal lines connect the minimum and maximum importance ratings for each task. Tasks include: (1) Lectures: attend lectures, seminars, and laboratory sessions; (2) Readings: complete assigned readings from textbooks and academic journals; (3) Exam prep: prepare for exams and assessments by reviewing material and creating study aids; (4) Writing: write research papers, essays, and reports; (5) Notes: take notes during lectures and review sessions; (6) Research: search for, evaluate, and synthesize academic sources; (7) Group projects: collaborate with peers on team assignments; (8) Office hours: meet with professors or TAs to seek clarification; (9) Revising: revise and edit written work based on feedback; (10) Campus resources: utilize tutoring centers, writing labs, and academic support services; (11) Labs/hands-on: complete laboratory experiments, field work, or studio projects; (12) Problem sets: solve homework exercises and practice problems; (13) Recordings: watch recorded lectures and instructional videos. Sample includes all task observations with non-missing importance data.

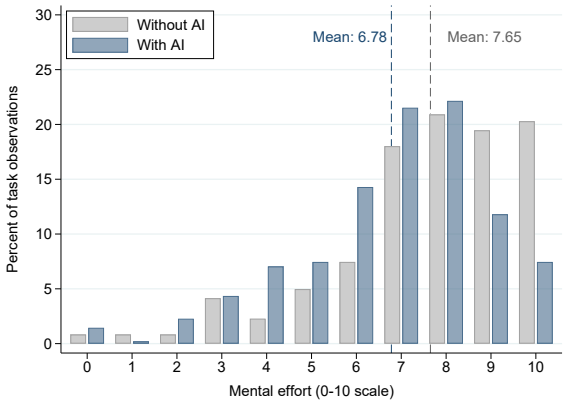
Figure A3: AI Adoption and Task Characteristics



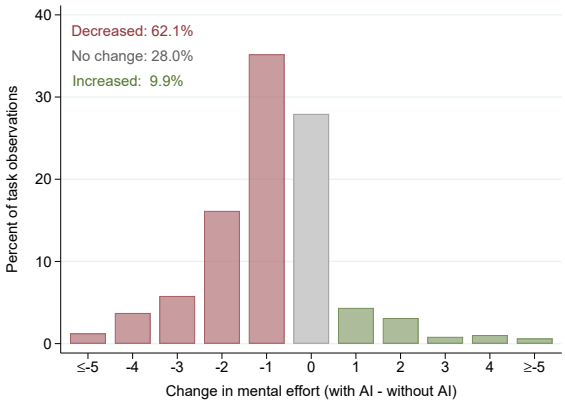
Notes: Each point represents one of 13 academic tasks. The y-axis shows the share of students using AI for each task. Panel A plots AI adoption against mean time spent on the task. Panel B plots AI adoption against the mean importance index, which averages ratings across all four importance dimensions. Panels C–F plot AI adoption against each importance dimension separately: learning, grades, skill development, and enjoyment. Dashed lines show linear fit.

Figure A4: Distribution of Mental Effort With vs. Without AI

Panel A. Effort Levels



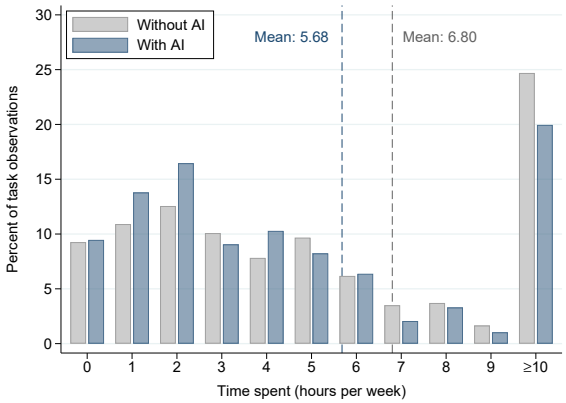
Panel B. Change in Effort



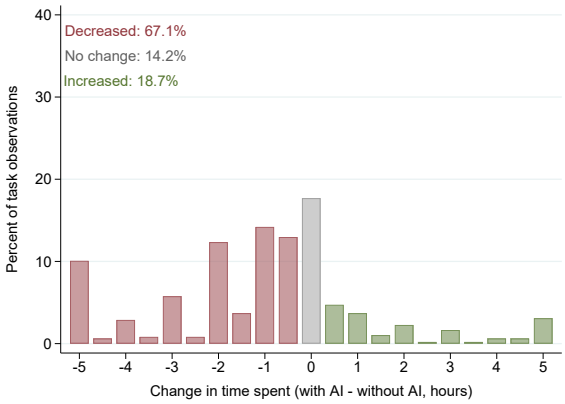
Notes: This figure shows the distribution of self-reported mental effort. Panel A compares effort with AI (navy) and without AI (gray, counterfactual). Panel B shows the distribution of within-task changes in effort (with AI minus without AI); green bars indicate decreased effort, gray bars indicate no change, and red bars indicate increased effort. The sample includes only AI users with non-missing effort data for both conditions. Dashed vertical lines indicate means.

Figure A5: Distribution of Time Spent With vs. Without AI

Panel A. Time Levels

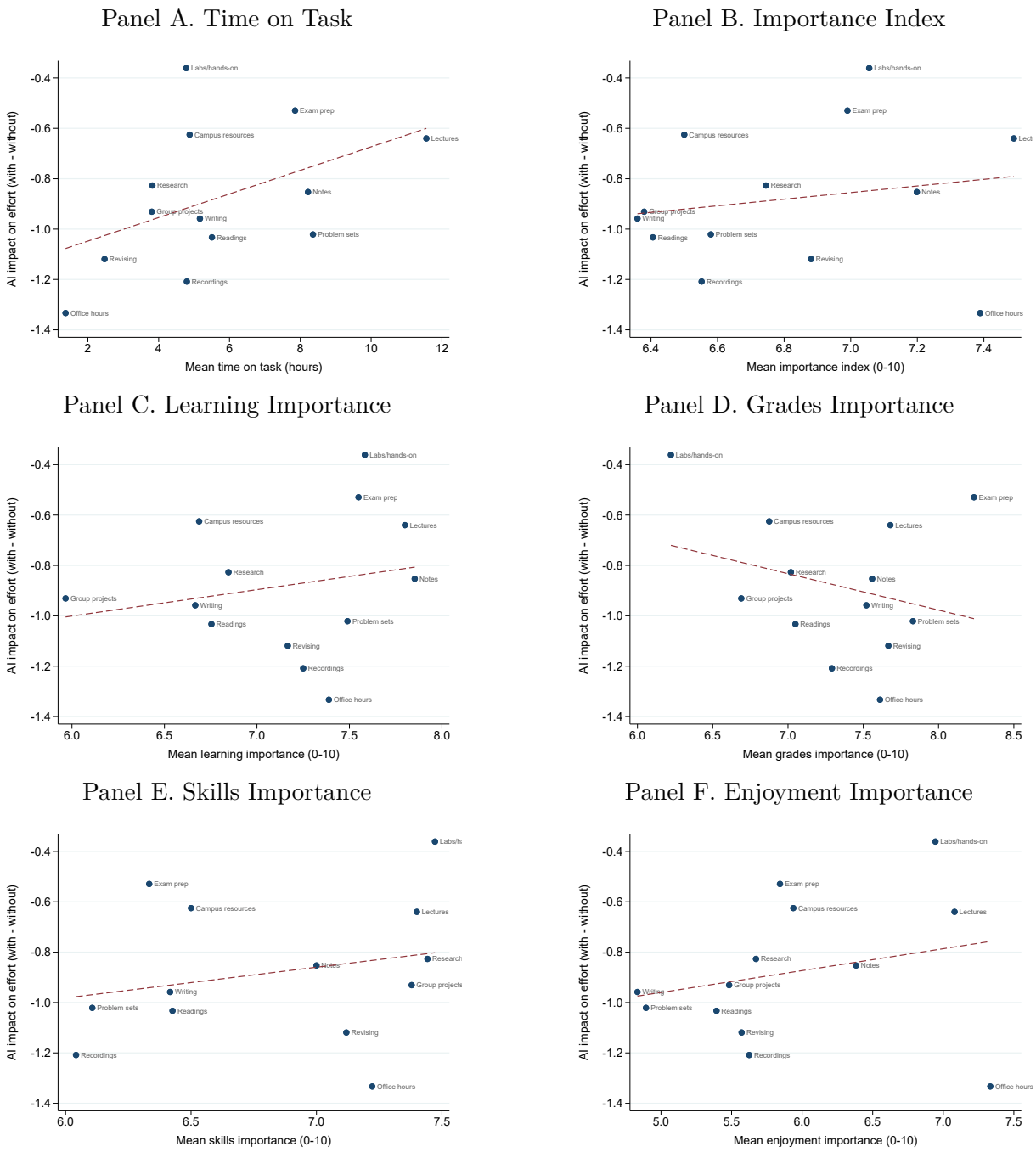


Panel B. Change in Time



Notes: This figure shows the distribution of self-reported time spent on academic tasks. Panel A compares time with AI (navy) and without AI (gray, counterfactual). Panel B shows the distribution of within-task changes in time (with AI minus without AI); red bars indicate decreased time, gray bars indicate no change, and green bars indicate increased time. The sample includes only AI users with non-missing time data for both conditions. Dashed vertical lines indicate means. Time is measured in hours per week.

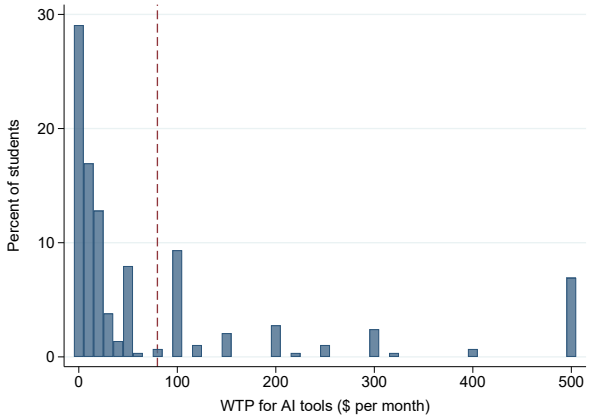
Figure A6: AI Impact on Critical Thinking and Task Characteristics



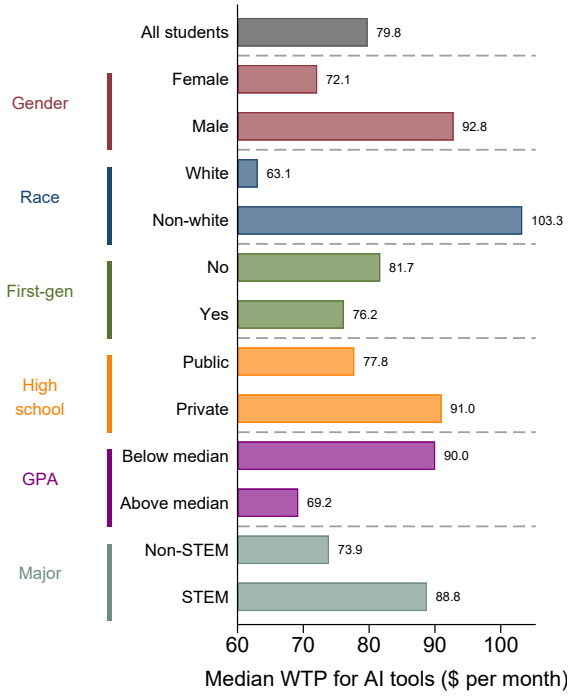
Notes: Each point represents one of 13 academic tasks. The y-axis shows the mean change in self-reported mental effort due to AI (effort with AI minus effort without AI), where negative values indicate AI reduces cognitive effort. Panel A plots effort change against mean time spent on the task. Panel B plots effort change against the mean importance index, which averages ratings across all four importance dimensions. Panels C–F plot effort change against each importance dimension separately: learning, grades, skill development, and enjoyment. Dashed lines show linear fit. The horizontal dashed line at zero indicates no change in effort.

Figure A7: Willingness to Pay for AI Tools

Panel A. Distribution of WTP

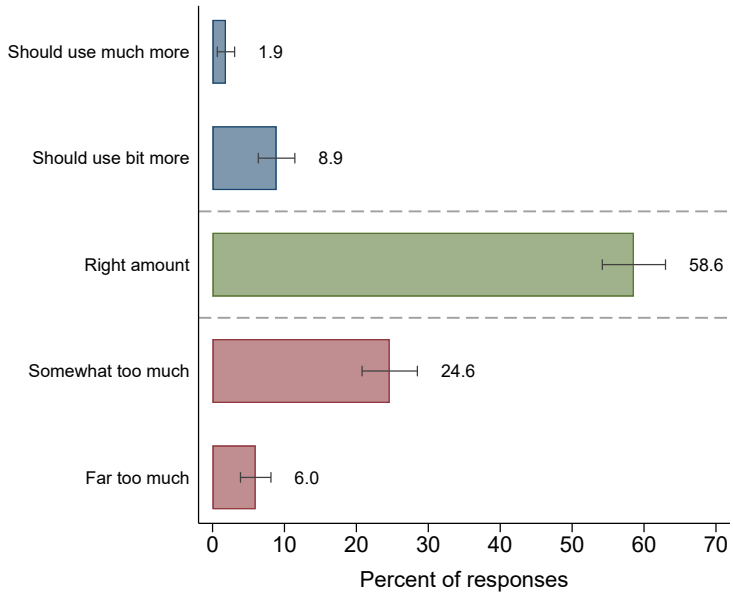


Panel B. WTP by Student Characteristics



Notes: This figure shows students' willingness to pay (WTP) for AI tools in dollars per month. Panel A displays the distribution of WTP, winsorized at the 95th percentile. The dashed vertical line indicates the mean. Panel B shows mean WTP by student characteristics. Error bars represent 95 percent confidence intervals. "First-gen" indicates first-generation college student status. GPA splits students at the median.

Figure A8: Do Students Use AI the Right Amount?



Notes: This figure shows students' self-assessment of whether they use AI the right amount for their academic tasks. The sample includes only AI users with non-missing self-assessment data. Maroon bars indicate students who feel they use AI too much; green bars indicate those who feel they use the right amount; navy bars indicate those who feel they should use AI more.

Table A1: Academic Tasks in Survey

Category	**No.**	**Description**
Content Consumption	1	Attend lectures, seminars, and laboratory sessions to receive instruction on course material and academic subjects.
	2	Complete assigned readings from textbooks, academic journals, and other course materials to prepare for class discussions and exams.
	3	Watch recorded lectures, instructional videos, or online course modules to learn course material.
Active Learning	4	Take notes during lectures and review sessions to capture key concepts and information for later study.
	5	Prepare for exams, quizzes, and assessments by reviewing course material and creating study aids such as flashcards, study guides, or concept summaries.
	6	Solve problem sets, homework exercises, and practice problems to apply course concepts and methods.
Knowledge Production and Research	7	Write research papers, essays, and reports applying academic writing conventions.
	8	Revise and edit written work based on feedback from instructors to improve academic writing skills.
	9	Search for, evaluate, and synthesize academic sources (journal articles, books, databases) to support assignments and research.
Collaboration and Support-Seeking	10	Collaborate with peers on group projects and team assignments to complete shared deliverables.
	11	Meet with professors, teaching assistants, or academic advisors during office hours to seek clarification on course material or discuss academic progress.
	12	Utilize campus resources such as tutoring centers, writing labs, and academic support services to enhance learning.
	13	Complete hands-on learning activities such as laboratory experiments, field work, practical exercises, or studio projects.

Notes: This table lists the 13 academic tasks included in the survey, grouped by category. Each participant responds to questions about three randomly selected tasks from this list. Tasks span four categories: content consumption (3 tasks), active learning (3 tasks), knowledge production and research (3 tasks), collaboration and support-seeking (4 tasks).